# ML19 Project data and a safe plan

Week 7, Your project | Session 19 of 24 | Machine Learning | New library: pandas

**Today's question.** Which dataset and which question can I answer with a fair test in two weeks?

**Objective.** By the end: you can describe a supplied dataset, spot a leakage column, and write a seven-heading project plan the instructor can approve.

Before this session: ML18 Checkpoint two with a locked test. If you cannot explain the last example, use its error card with the instructor first.

## Words for today

| Word | What it means | Think of it like this |
|---|---|---|
| project plan | A short page that fixes the question, data, target, inputs, baseline, test and limits before any building. | A drawing of the house before laying bricks. |
| leakage column | A column only known after the answer, so it must not be an input. | A tip that is only known after the ride. |
| inputs | The columns known before the prediction. | The clues you are given. |
| limits | What the data and the model cannot tell us. | The edges of the map. |

## Block A. Tour the fares data

synthetic_fares.csv holds 150 invented boda-boda trips. Look at its shape and at how each column moves with the fare.

**Predict.** Predict which column moves most closely with fare_ugx. Write your guess on paper first.

*Your guess, written before you run it*

```
import pandas as pd
fares = pd.read_csv("synthetic_fares.csv")
print(fares.shape)
corr = fares[["distance_km", "rush_hour", "raining", "tip_ugx",
    "fare_ugx"]].corr()["fare_ugx"].round(2)
print(corr)
```

**Run it, then compare.** The table has 150 rows and 6 columns. Distance has a correlation of 0.76 with the fare, rain 0.12 and rush hour about minus 0.08. The tip column has the highest value, 0.88. A correlation near 1 means the two columns rise together. The tip rises with the fare because it is a share of the fare, and it is only known after the trip, so using it to predict the fare would be cheating. That makes tip_ugx a leakage column.

**Investigate.**

1. Why can tip_ugx not be an input?

2. Which two columns are sensible inputs?

**Modify.** Print the correlation of the fare with the trip_id column.

## Block B. A plan needs seven headings

The plan has seven headings: question, dataset, target, inputs, baseline, evaluation and limits. Write one sentence under each.

**Predict.** Write the target and two inputs for your project before looking at the example. Write your guess on paper first.

*Your guess, written before you run it*

**Run it, then compare.** An example plan: question, can distance and rush hour predict a boda-boda fare; dataset, synthetic_fares.csv; target, fare_ugx; inputs, distance_km, rush_hour and raining; baseline, the mean fare of the training trips; evaluation, mean absolute error in shillings on a locked test; limits, invented data and 150 trips. The options are synthetic_homes.csv (rent from area and rooms), synthetic_fares.csv and the iris flowers. Real people's records are not used. Repeat the AI rules: any assisted step is logged in AI_USE_LOG.md after your own attempt, or the log says No AI assistance used.

**Investigate.**

1. Which heading stops you using the test answers to choose a model?
2. Why is the home_id column not an input for homes?

**Modify.** Write the leakage column you will exclude and why.

## Make

Write PROJECT_PLAN.pdf using the seven headings. Name one leakage column and exclude it. Ask the instructor to approve the scope before you build.

Save your work as ML19. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

## Error card

| You see | It means | Fix |
|---|---|---|
| KeyError: 'tip' | The column is called tip_ugx. | Print fares.columns and copy the exact names. |
| The plan promises too much | It asks for real people's data, an app or a neural network. | Reduce it to one supplied table, one target and one baseline. |

## Exit ticket

A project plan names the question, target, inputs and test before any code, and it excludes the column that gives the answer away.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.